



United International University
Department of Computer Science and Engineering
CSE 1325: Digital Logic Design, Final Exam, Spring 2026
Total Marks: 40, Duration: 2 hours

[Any examinee found adopting unfair means including copy from another examinee will be expelled from the trimester/program as per UIU disciplinary rules.]

Answer All Questions

1. A smart security gate monitors a serial binary data stream coming from an access sensor. The gate is programmed to recognize a specific binary pattern. Whenever the pattern is detected in the incoming data stream, the system generates an output of 1; otherwise, the output remains 0. For the input sequence,
 $x = 010100010101111$, the corresponding output sequence produced by the system is
 $y = 000100000101000$.

The circuit is implemented using D flipflops.

- Determine the binary pattern recognized by the gate circuit. [1]
 - Draw the state diagram of the circuit and assign states. [2]
 - Draw the state table with output and flip flop inputs. [2]
 - Find the minimized the functions of output and flip flop inputs. [2]
 - Draw the circuit diagram using flip flops and basic gates. [1]
2. The BSS Enterprise's automated control system continuously monitors five types of alerts. These alerts may occur simultaneously, but Commander Data can only respond to one alert at a time. A 5-input Priority Encoder is used to determine which alert should be handled first. The alert types, their input signals, and their output codes are listed below:

Priority	Alert Type	Input - Output
1 (highest)	Hull Breach Detected	B – 111
2	Reactor Failure	R – 101
3	Oxygen Level Critical	O – 011
4	Communication Request	C – 010
5 (lowest)	Routine System Report	S – 001

The encoder also produces a Valid (V) output.

Provide the following:

- Derive the truth table of the priority encoder, including the valid bit. [3]
- Derive the Boolean expressions for all the outputs. [3]
- Draw the logic diagram using basic gates. [2]

Q(t)	Q(t+1)	D	Operation
0	0	0	Reset
0	1	1	Set
1	0	0	Reset
1	1	1	Set

3. a) A basic digital counting mechanism is incorporated within a system to facilitate the automatic decrementing of stored values (for instance, tracking the number of remaining items). Design a 4-bit decrementing ripple counter using D flip-flops that respond to changes occurring at the instant when the clock signal transitions from a lower level to a higher level. [4]

- b) A digital system needs a register that can perform different operations based on control signals. Design a 4-bit universal shift register using D flip-flops with control inputs A and B such that: [4]

A = 0, B = 1 → The bit positions are arranged in the opposite order

A = 1, B = 0 → Each bit takes the complement of its earlier state

A = 0, B = 0 → The contents move one position toward the left

A = 1, B = 1 → All bits are cleared

4. A smart greenhouse uses four sensors to control an automatic cooling system:

- A = Temperature is high
- B = Humidity is high
- C = CO₂ level is high
- D = Sunlight intensity is high

The cooling system should turn ON when any of the following conditions are true:

- Temperature is high and humidity is high.
- Temperature is high and sunlight intensity is high.
- CO₂ level is high while humidity is low.

As a digital system designer you need to use a 4:1 MUX and any other basic gates. Do the following:

- i. Obtain the truth table [2]
- ii. Find the optimized MUX input equations [3]
- iii. Draw the complete circuit diagram [3]

5. It is the FIFA World Cup 2026 Final. Cristiano Ronaldo is one goal away from winning the World Cup for his country and from an unthinkable milestone, his 1000th career goal. He slots it into the net and the stadium erupts but VAR intervenes.

Three offside cameras A, B and C each output 1 if they confirm Ronaldo was onside, and 0 otherwise. The goal is awarded ($F = 1$) only when the majority of cameras agree he was onside. The VAR control room has no 3×8 decoder. They only have 2×4 decoders with Active High Enable and Active Low Output. As a die-hard CR7 fan in the VAR booth, you must build the system yourself using the 2×4 decoders. The world is watching!

- i. Complete the truth table for the goal confirmation system with inputs A, B, C and output F. [3]
- ii. Find the Canonical SOP form for the goal confirmation function F
- iii. Using exactly the two 2×4 decoders and a minimal number of logic gates, implement the function F. Draw the complete circuit diagram, clearly labeling all inputs, enable lines, minterm outputs, and gate connections to produce F. [1] [4]